

Intrinsic-Signal “Do-No-Harm” Gating for a Frozen-Base Soft-Prompt Memory Wrapper

Settings, Metric Definitions, and a Signal–Usefulness Correlation Study (v1.5)

Plan 08 (mem-X line) — internal technical note

2026-06-04

Abstract

v1 of this project showed that a learned K -token soft-prompt memory wrapper around a frozen LLM is a *narrow lossy compressor*: it matches the Gist-Tokens baseline on QuALITY but *negatively transfers* on benchmarks it never trained for (verbatim retrieval, multi-hop QA), sometimes scoring *below* the base model with no wrapper at all. Because the whole point of a frozen base is that adding a module should not *lose* points, v1.5 asks: can we *gate* the wrapper’s intervention — suppress it when it would hurt — using cheap signals computable at inference? This note fixes **all experimental settings** (§3) and **all metric definitions** (§4), both the benchmark-native black-box outcomes and the intrinsic signals (including *per-layer* logit-lens quantities), then reports the first correlation study (§5) identifying *which* intrinsic signals — and at *which layers* — predict the benchmark’s own usefulness measure. The design principle: first learn what correlates with the target, then design the gate around it.

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1 Deliverables and source links

This PDF is the compiled v1.5 technical note. The links below are the source-of-truth entry points for the deliverables, facts, and main result summaries around Plan 08.

1.1 v1 wrapper facts and main result

- **Settings registry:** [settings.md](#). Every result cell in this report should trace to a setting ID; v1 headline cells use **P08-S2**, and v1.5 signal-grid cells use **P08-S3**.
- **Main v1 result summary:** [v1-results-2026-06-03.md](#). This is the main facts document for the mem-embedding wrapper, including the three-regime transfer law.
- **v1 pivot / lessons:** [v1-if-wrapper-doesnt-work-2026-06-03.md](#). This explains why v1 is framed as a characterization paper and how it leads to v1.5.
- **History / chronology:** [v0-how-we-got-here.md](#).

1.2 v1.5 signal grid and gating deliverables

- **Compiled report:** [v1.5-intrinsic-gating-study-2026-06-04.pdf](#).
- **LaTeX source:** [v1.5-intrinsic-gating-study-2026-06-04.tex](#).
- **Signal-grid folder:** [grids-2026-06-04](#).
- **Main signal-ranking figures:** [rank_help.png](#) and [rank_interesting.png](#).
- **Signal-grid CSVs:** [grid_auroc_correct.csv](#), [grid_auroc_noharm.csv](#), [grid_spearman_cont.csv](#), [corr_long.csv](#).

1.3 v2 and idea table

- **v2 plan:** [v2-plan.md](#).
- **v2 related work:** [v2-related-work.md](#).
- **Q2 activation-memory probe:** [q2-activation-memory-probe.md](#).
- **Plan table:** [notes/plans/README.md](#).
- **Full idea table:** [notes/ideas/README.md](#), including H6 (Plan 08) and related follow-ups H3/H7/G4/G5.
- **Original brainstorm:** [inference-time-training.md](#).

2 Experiments: what was run, what is running, what is planned

2.1 Compute environment

Three Kubernetes pods (shared, multi-tenant): **sam-dev-test** (1×H200-143GB), **sam-dev-ray** (4×H100-80GB), **sam-dev** (4×H100-80GB). Qwen3-8B is present on all three; Qwen3-14B only on **sam-dev**. Each probe run trains one wrapper (one seed) on a single GPU and evaluates all assigned benchmarks; seeds are sharded one-per-GPU to use the cluster in parallel. All runs use the memory-safe probe (`scripts/probe_v2.py`): last-position hidden states are cloned and the full activation tensors freed, a single encoder pass is taken per chunk, and the CUDA cache is flushed periodically — a fix introduced after an out-of-memory postmortem on the 80 GB cards (retained all-position `hidden_states` views + redundant encoder passes).

2.2 Runs ledger

run	setting	purpose	seeds	n/bench	benchmarks	status
Phase 0	P08-S3	prediction-derived signals, free-gen scored	4–5	800	QuALITY, RULER, Trivia, MARCO, MuSR, Hotpot, SQuAD,	done (§5.1)
seed-42 probe	P08-S3	native-MC + 37-layer signals (single seed)	1	150	QuALITY, RULER, Trivia, MuSR, Hotpot,	done (§5.2)
9-seed extended	P08-S3	full battery, multi-seed (robust)	9	1350	QuALITY, RULER, Trivia, MuSR, Hotpot,	done (§5.4)
new-bench (8B)	P08-S3	4 more answer-type angles	5	150	QuALITY-hard, SQuAD-v2, Nar-rativeQA, MS-MARCO	running (seed42 done)
14B cross-scale gate (Phase 4)	P08-S3	does the signal hold at 2× scale?	—	—	main 5	deferred (pod busy)
	P08-S4	input-conditioned do-no-harm gate	—	—	public-only	planned (§8)

The robust conclusions (§5.4) rest on the *9-seed extended* run; Phase 0 and seed-42 are earlier, smaller passes kept for traceability and because seed-42 illustrates the granularity effect (and one single-seed artifact the 9-seed run corrects).

2.3 Probe protocol (per eval item)

On the frozen base we run: (i) the **wrapper** (memory prefix + query) with hidden-state output; (ii) **no_context** (query only); (iii) a **memory-ablation** pass (wrapper prompt with attention to the K memory tokens masked) for the causal-influence signal; (iv) greedy generation for wrapper and **no_context**; (v) for MC benchmarks, per-option log-likelihood scoring; (vi) a teacher-forced pass over the wrapper’s own generated answer for answer-span statistics. From these we log the native black-box outcome (§4.1) and the full intrinsic battery across granularities and angles (§4.3) — ~ 6 – 10 forward passes per item, ~ 50 GPU-min per seed for five benchmarks at $n=150$.

3 Settings

Unless a table states otherwise, numeric signal/correlation results in this report use setting [P08-S3](#). The underlying wrapper recipe is [P08-S1](#); v1 three-regime background comparisons use [P08-S2](#).

3.1 Frozen base and the wrapper

The base f_θ is **Qwen3-8B** (hidden size $d = 4096$, 36 decoder layers), *frozen* throughout; only the wrapper trains. The wrapper ϕ_ψ (≈ 218 M trainable params, 2.67% of the base) maintains a memory state $m_t \in \mathbb{R}^{K \times d}$, $K = 32$.

Write side (recurrence). For each context chunk c_t ($t = 1:N$, $N = 6$ chunks of ≤ 256 tokens), a single Perceiver-IO cross-attention+FFN block g_ϕ (heads $h = 8$, FFN mult 2, $L = 1$ layer) produces a candidate, and the wrapper exposes the residual:

$$\Delta m_t = \text{LN}(g_\phi(m_{t-1}, c_t) - m_{t-1}), \quad m_t = m_{t-1} + \alpha_{\text{upd}} \cdot \Delta m_t, \quad (1)$$

with α_{upd} a learned scalar (init 1.0). Δm_t is published on `MemoryState.extra["last_delta"]` for geometry probes.

Read side (combine). At query time the final memory m_T is delivered to the base. The canonical mode is **xattn**: a second cross-attention lets memory re-read the raw chunk pool H_{chunks} , then the refined memory is prepended as a soft prefix:

$$m'_T = m_T + \alpha_{\text{ro}} \cdot \text{LN}(\text{CrossAttn}(m_T, H_{\text{chunks}})), \quad \text{inputs_embeds} = [m'_T; x], \quad (2)$$

where x are the query-token embeddings. Seven combine modes exist (**prefix**, **residual**, **interleave**, **hybrid**, **xattn**, **xattn_residual**, **gated**); only **gated** is unimplemented and is the *target* of v1.5 (§8). No combine mode ever moves a base weight.

Answer head (auxiliary, training only). For categorical training tasks a `Linear( $d, n_{\text{cls}}$ )` head on `pool( $m_T$ )` supplies a direct class-prediction gradient that bypasses the frozen base.

Joint training loss.

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{kl}} \text{KL}(p_{\text{student}} \| p_{\text{teacher}}) + \lambda_{\text{div}} \frac{1}{T-1} \sum_t \cos(\Delta m_t, \Delta m_{t-1}) + \lambda_{\text{ah}} \mathcal{L}_{\text{ah}}, \quad (3)$$

teacher = same base reading the full context (**full_context**).

3.2 Training recipe (locked)

knob	value	knob	value
base	Qwen3-8B (frozen, bf16)	combine	xattn
K (memory tokens)	32	h heads / L layers	8 / 1
train dataset	categorical_niah	N train items	500
N chunks	6	max chunk tokens	256
optimiser	AdamW, lr 3×10^{-5} , const	steps	1800 (SFT)
λ_{kl}	1.0	λ_{div}	0.1
λ_{ah} (answer head)	0.5	seeds	{42,7,11,13,1,5,17,99,23}

3.3 Evaluation protocol

The wrapper is trained *only* on **categorical_niah** and evaluated **zero-shot** on public benchmarks via a `--eval-dataset` flag (wrapper weights never updated on eval data). For every eval item we run, on the same frozen base: (i) the **wrapper** (memory prefix + query); and (ii) **no_context** (query only) — which *is* the base *without* the wrapper, since the wrapper replaces the document with K memory tokens. **no_context** is therefore the do-no-harm reference (§4.2). We also have the standard baselines **full_context** (base reads the full document), **retrieval** (BM25 top-2 chunks), **summary**.

Compute fan-out. The probe (`scripts/probe_v2.py`) trains one wrapper per seed and evaluates all 5 benchmarks with full instrumentation. Seeds are sharded one-per-GPU across 8 GPUs (`sam-dev-test` H200 $\times 1$; `sam-dev-ray` H100 $\times 4$; `sam-dev` H100 $\times 4$) to produce a multi-seed dataset.

3.4 Benchmarks

All are public long-context QA, evaluated zero-shot. “Answer space” drives the native metric (§4.1).

benchmark	source	answer space	n_{eval}	native metric
QuALITY	Pang et al. 2022	4-way MC (A–D)	150	accuracy (likelihood)
MuSR-mm	Sprague et al. 2024	MC (murder myst.)	150	accuracy (likelihood)
RULER-NIAH	Hsieh et al. 2024	6-digit value	150	contains-recall
HotpotQA	Yang et al. 2018	free-form span	150	SQuAD EM/F1
TriviaQA-RC	Joshi et al. 2017	span + aliases	150	EM/F1 (max over aliases)
SQuAD v2	Rajpurkar et al. 2018	span / no-answer	—	EM/F1 (+abstain)
NarrativeQA	Kočiský et al. 2018	open-ended	—	ROUGE-L, BLEU
MS MARCO	Bajaj et al. 2016	free-form	—	ROUGE-L, BLEU-1

4 Metric definitions

We separate three families: (a) the benchmark-*native* black-box *outcome* metrics (§4.1); (b) the *usefulness* labels derived from them, which are what a gate should track (§4.2); and (c) the *intrinsic signals* that are candidate gate *inputs* (§4.3).

4.1 Black-box (benchmark-native) outcome metrics

Principle: score by the benchmark’s own method. Notably, the multiple-choice benchmarks are natively scored by *likelihood*, not free-form generation; our earlier free-generation + letter-

parsing under-counted the base (e.g. `full_context` appeared as 0.079 on QuALITY purely as a parsing artifact).

(MC) Length-normalised option log-likelihood. For options $o^{(1)}, \dots, o^{(C)}$ with token sequences $o = (o_1, \dots, o_{|o|})$, score each by the mean log-prob of its continuation given the prompt P :

$$s(o) = \frac{1}{|o|} \sum_{i=1}^{|o|} \log p_{\theta}(o_i \mid P, o_{<i}), \quad \hat{o} = \arg \max_o s(o), \quad \text{acc} = \mathbf{1}[\hat{o} = o^*]. \quad (4)$$

We also record the continuous *margin* `mc_margin` = $s(o^*) - \max_{o \neq o^*} s(o)$ and the gold log-prob $s(o^*)$ — richer (less noisy) targets than binary accuracy at $n=150$.

(span) SQuAD EM / F1. With SQuAD normalisation $\mathcal{N}(\cdot)$ (lowercase; strip punctuation; drop articles {a,an,the}; collapse whitespace):

$$\text{EM} = \mathbf{1}[\mathcal{N}(\hat{y}) = \mathcal{N}(y)], \quad \text{F1} = \frac{2 \text{prec} \cdot \text{rec}}{\text{prec} + \text{rec}}, \quad (5)$$

on the multiset of normalised tokens; over a reference/alias set R the score is $\max_{y \in R}$ (TriviaQA aliases, NarrativeQA refs). HotpotQA sup-fact/joint metrics are *not* applicable (the wrapper predicts no supporting facts). SQuAD v2: a no-answer item scores 1 iff the model abstains; our current harness restricts to answerable items (SQuAD-1.1-style) — an open gap.

(retrieval) RULER contains-recall. `em_value` = $\mathbf{1}[y \subseteq \hat{y}]$ (the planted 6-digit value appears in the output; official `acc_match_str_any`).

(generation) ROUGE-L / BLEU. ROUGE-L is the LCS-based F-score $\frac{2P_{\text{lcs}}R_{\text{lcs}}}{P_{\text{lcs}}+R_{\text{lcs}}}$, $P_{\text{lcs}} = \text{LCS}(\hat{y}, y)/|\hat{y}|$, $R_{\text{lcs}} = \text{LCS}(\hat{y}, y)/|y|$; BLEU- n is the brevity-penalised geometric mean of clipped n -gram precisions. Both taken max/over the reference set. ROUGE-L is the reported headline for NarrativeQA and MS MARCO.

4.2 Usefulness labels (the gate’s target) and reference

Let u_w be the wrapper’s native correctness on an item and u_b the reference’s. We define per item:

$$\begin{aligned} \text{correct} &= \mathbf{1}[u_w \geq \tau] \quad (\tau=0.5), \\ \text{noharm} &= \mathbf{1}[u_w \geq u_b], \\ \text{help} &= \mathbf{1}[u_w > \tau \wedge u_b \leq \tau], \quad \text{hurt} = \mathbf{1}[u_w \leq \tau \wedge u_b > \tau], \\ \text{cont} &= \text{continuous target (MC: mc_margin; else native value)}. \end{aligned}$$

Two references. Against `no_context` (base without wrapper) the wrapper is already harmless on 80–100% of items per benchmark (§5); the damaging gap is against `full_context` (compression discards usable context). Hence the gate is best understood as a *compressibility detector*: decide when lossy memory is safe vs. when to fall back to more context. The *ideal* gate maximises `noharm` (never drop below the base) while keeping `help`.

4.3 Intrinsic signals (candidate gate inputs)

All are computed at inference without the gold label. Let the wrapper run produce final-token next-token distribution p_w and per-layer last-token hidden states h_ℓ^w ($\ell = 0:36$); the `no_context` run gives p_0, h_ℓ^0 .

Final-layer signals. Entropy $H(p) = -\sum_v p_v \log p_v$; confidence $\text{conf} = \max_v p_v$; margin $= p_{(1)} - p_{(2)}$ (top-two gap); divergence $\text{kl_w0} = \text{KL}(p_w \| p_0)$; agreement $\mathbf{1}[\arg \max p_w = \arg \max p_0]$. Reported as `entropy_w, conf_w, margin_w, kl_w0, first_tok_agree`.

Per-layer logit-lens signals (the multi-layer probe). Project each layer’s last-token hidden through the frozen final norm and unembedding W_U to get an early “readout” distribution:

$$\pi_\ell = \text{softmax}(W_U \text{LN}_f(h_\ell)). \quad (6)$$

From π_ℓ^w we record `ll_entropy_w_l, ll_conf_w_l, ll_margin_w_l`; the wrapper-vs-base divergence `ll_kl_wb_l` $= \text{KL}(\pi_\ell^w \| \pi_\ell^0)$; and the depth-stabilisation flag `ll_argmax_agree_final_l` $= \mathbf{1}[\arg \max \pi_\ell^w = \arg \max p_w]$. We also record per-layer *hidden drift*

$$\text{drift_l} = \frac{\|h_\ell^w - h_\ell^0\|_2}{\|h_\ell^0\|_2 + \epsilon}, \quad (7)$$

i.e. how much the wrapper perturbs each layer’s residual stream at the answer position. **Rationale (the multi-layer hypothesis):** different depths encode different things — early-layer drift may flag format perturbation (harmful), mid-layer π_ℓ agreement-with-base may flag safe compression, late-layer confidence may flag correctness — so the most predictive signal need not live at the final layer, and a *per-layer* gate can read whichever layer carries the signal.

Memory geometry. `drift_mem` $= \|m_T - m_0\|_F$; `delta_mean` $= \frac{1}{N} \sum_t \|\Delta m_t\|$; `q_chunk_cos` $= \cos(\overline{\text{enc}}(q), \overline{\text{enc}}(c))$ (query vs. chunk mean-pooled encodings; max and mean over chunks).

Ensemble signals (Phase 0 only; not purely intrinsic). `intervene_vs_noctx` $= \mathbf{1}[\hat{y}_w \neq \hat{y}_0]$; `agree_retrieval` $= \mathbf{1}[\hat{y}_w = \hat{y}_{\text{retr}}]$; `n_baselines_agree` $= \#\{\text{baselines whose prediction} = \hat{y}_w\}$; `w_outlen` (generated length). These need running other strategies and so are candidate *ensemble* gate inputs.

4.4 Correlation metrics

For each (signal, layer) we report **AUROC** against a binary label (rank statistic: $\text{AUROC} = \Pr[s_+ > s_-]$; $0.5 = \text{no signal}$, $|\cdot - 0.5|$ measures discriminative power, < 0.5 means the signal is *inversely* predictive), and **Spearman** ρ against the continuous target. A *per-layer profile* reports, per layer-metric, the layer maximising $|\text{AUROC} - 0.5|$. A planned multivariate logistic probe on the full {metric×layer} vector estimates the achievable gate ceiling.

5 Results and interpretation

Setting for all result cells in this section: [P08-S3](#), unless explicitly labeled as v1 three-regime background ([P08-S2](#)).

Status. The **9-seed multi-seed study** ($n=1350/\text{bench}$) is **complete** and is the robust result (§5.4); it *supersedes* the single-seed seed-42 rankings (§5.2) and corrects one of them. Seed-42 tables are kept for traceability and because they illustrate the granularity effect.

5.1 Phase 0 — base rates (free-gen scored, 5,600 items)

Setting for every cell in this table: P08-S3.

bench	u_w	no_context	full_context	help%	hurt%	noharm%
quality	0.193	0.141	0.079	13	8	92
musr_mm	0.492	0.494	0.551	11	12	89
trivia_qa	0.168	0.338	0.565	3	20	80
hotpot_qa	0.086	0.104	0.438	3	5	95
ruler_niah	0.000	0.000	0.994	0	0	100

Read: vs. **no_context** the wrapper is mostly harmless (noharm 80–100%); it *actively* hurts only on TriviaQA (20%). The catastrophe is vs. **full_context** (RULER 0.994→0, Hotpot 0.44→0.09) — compression discards usable context. So the gate is a compressibility detector, not a corruption filter.

5.2 Seed-42 gate signal sheet (native-MC + per-layer)

Setting for every cell in this subsection: P08-S3 (single-seed slice).

bench	native metric	best intrinsic signal	AUROC	predictive layer
hotpot_qa	squad_f1	conf_w	0.753	near-final (34–35)
quality	accuracy (LL)	11_kl_wb@30	0.680	mid-layer 30
trivia_qa	squad_f1	11_conf@34	0.653	layer 34
musr_mm	accuracy (LL)	11_entropy@11	0.582	weak (at chance)
ruler_niah	exact_value	—	—	undefined (0 correct)

Per-layer profile (layer maximising $|\text{AUROC} - 0.5|$ for **correct**; bold = best per row):

signal	quality	hotpot	trivia	musr
11_kl_wb_l	@30: 0.680	@34: 0.332	@33: 0.284	@34: 0.555
11_conf_w_l	@30: 0.353	@35: 0.731	@34: 0.653	@22: 0.428
11_margin_w_l	@30: 0.333	@35: 0.696	@34: 0.630	@19: 0.422
drift_l	@2: 0.374	@33: 0.397	@26: 0.370	@17: 0.531
11_argmax_agree_final_l	@34: 0.461	@34: 0.597	@23: 0.343	@34: 0.479

5.3 Interpretation

1. **The wrapper’s own confidence/margin predicts correctness** on the QA benchmarks (conf_w/11_conf AUROC 0.65–0.75). A confidence-thresholded gate is therefore viable: low confidence $\Rightarrow$ suppress the wrapper.
2. **The predictive layer is benchmark-dependent and often not the final one.** QuALITY’s strongest signal is logit-lens KL-vs-base at *layer 30* (11_kl_wb@30 = 0.680), whereas Hotpot/Trivia peak at layers 34–35. This empirically validates the multi-layer hypothesis: a single-layer (output-only) gate would miss the QuALITY signal; a per-layer gate has the right information to read.

3. **Where there is no signal, the gate should simply close.** MuSR is at chance (max AUROC 0.58) and RULER is uniformly wrong (AUROC undefined). These are exactly the regimes where the wrapper must defer to context; a do-no-harm gate that closes there is correct, not a failure.
4. **Ensemble signals are stronger but costlier** (Phase 0: `n_baselines_agree` AUROC 0.83–0.90 on extractive benches), trading deployability for accuracy — a fallback if intrinsic signals prove insufficient under multi-seed.

5.4 Multi-seed result (9 seeds, $n=1350/\text{bench}$) — robust

Setting for every cell in this subsection: P08-S3 (robust 9-seed slice). Pooling seeds {42, 7, 11, 13, 1, 5, 17, 99, 23} with the full extended battery sharpens and *corrects* the single-seed picture.

bench	best signal	angle	AUROC	Spearman
trivia_qa	<code>seq_logprob</code>	confidence	0.769	+0.357
	<code>l1_kl_wb@34</code>	divergence	0.229 (inv.)	−0.40
hotpot_qa	<code>top5_mass_w</code>	confidence	0.761	+0.316
quality	(none robust)	—	≤ 0.55	—
musr_mm	(none)	—	≤ 0.52	—
ruler_niah	undefined	—	—	—

1. **Wrapper output confidence robustly predicts correctness** on extractive QA (`seq_logprob/top5_mass` AUROC 0.76–0.77); the cheapest granularity (output/span) wins, and per-layer `l1_conf` aggregates add only a little (trivia: final 0.60 → max 0.66).
2. **Wrapper→base divergence is inversely predictive** (`l1_kl_wb`, `kl_0w`, `js`, `tv`): a larger distribution shift $\Rightarrow$ the wrapper is more likely *wrong* (trivia `l1_kl_wb@34` $\rho=-0.40$). A natural gate input, used inversely.
3. **The single-seed QuALITY “layer-30 KL” peak (0.68, §5.2) was an artifact** — it falls to ≈ 0.47 at 9 seeds. No intrinsic signal robustly predicts QuALITY correctness; this is exactly why the multi-seed study was necessary.
4. **Where there is no signal (musr/ruler), the gate should close** (do no harm) — the correct behaviour, not a failure.

Gate input (decided): wrapper output *confidence* + wrapper→base *divergence* (inverse), both robust at $n=1350$; per-layer features are secondary inputs to the multivariate probe.

6 Signal codebook: setting, definition, and rationale

Setting for every correlation in this report. Each signal is computed in the per-item probe protocol of §2, on a *frozen* Qwen3-8B with the wrapper trained zero-shot on `categorical_niah` (§3; setting P08-S3). The rankings of §7 are pooled over **9,750 items across 9 benchmarks** (main five at 9 seeds, new four at 5 seeds); targets `help` (wrapper correct $\wedge$ base wrong), `noharm` (wrapper $\geq$ base) and `lift` are as in §4.2; AUROC/Spearman as in §4.4. “Pass” = which forward(s) the signal reads. AUROC<0.5 = *inverse* predictor.

6.1 Confidence / uncertainty angle (obvious — removed in §7)

signal	pass / setting	why it relates (and why it is obvious)
conf_w, margin_w, entropy_w, top5_mass_w	first generated-token distribution, wrapper run	model self-confidence; a sharp/low-entropy distribution $\Rightarrow$ correct. The “model is sure $\Rightarrow$ right” calibration effect. (top5_mass_w is the top help predictor, AUROC 0.653.)
span_entropy_*, span_conf_mean, seq_logprob, perplexity	teacher-forced pass over the wrapper’s own greedy answer	the same confidence, aggregated over the whole answer span rather than one token.
conf_0, entropy_0, margin_0	no_context run (base, query only)	<i>item difficulty</i> : if the base is confident with no context, the item is easy and the wrapper is unlikely to make it worse. (entropy_0 is the top noharm predictor, AUROC 0.646.)
ll_{conf,margin,entropy}_len	logit-lens of the wrapper hidden state, per layer	depth-resolved confidence.

6.2 Divergence-from-base angle (interesting)

signal	pass / setting	why it relates (non-obviously)
kl_w0, kl_0w, js_w0, tv_w0	first-token distribution, wrapper <i>vs</i> no_context	how far the memory moves the base’s <i>output</i> distribution; a large move means the wrapper is overriding the base’s prior, and empirically that override is usually wrong (inverse).
ll_kl_wb_l@l	logit-lens KL between wrapper and no_context hidden at layer ℓ	deviation of the base’s <i>internal</i> representation at depth ℓ . Mid-late layers ($\ell \approx 25$ – 34) are where the answer decision forms; large KL there = the memory is fighting the base’s own reasoning $\Rightarrow$ harm. The strongest non-obvious correlate.
mem_influence_kl, mem_influence_argmax_change	wrapper run <i>vs</i> the same no_context run with attention to the K memory tokens masked	causal: how much the base actually <i>uses</i> the memory. Double-edged (genuine use <i>vs.</i> wrong override).

6.3 Representation-geometry angle (interesting)

signal	pass / setting	why it relates (non-obviously)
drift_l0l, drift_l::late	$\ h_\ell^w - h_\ell^0\ / \ h_\ell^0\ $ at the answer position, per layer	raw magnitude by which the soft prompt perturbs the residual stream. A pure geometry quantity, yet mid-late perturbation predicts harm: gentle steering helps, violent distortion hurts.
drift_mem	$\ m_T - m_0\ _F$ of the recurrence	total movement of the memory state from its learned init; encodes “how much was written” but explodes on collapse.
delta_mean/max/last/trend	per chunk $\ \Delta m_t\ $ over the recurrence	shape of the write process; delta_trend (growing <i>vs.</i> decaying updates) distinguishes a recurrence that <i>integrates</i> from one that <i>chases</i> each chunk.
slot_norm_{mean,std,max}, slot_active_frac	slot magnitudes of m_T	how many of the K memory slots are actually used; a degenerate (few-active) memory encodes less.

6.4 Relevance and agreement angles

signal	pass / setting	why it relates
<code>q_chunk_{max,mean}_cos</code> <code>q_chunk_spread</code>	sine of mean-pooled query vs. chunk encodings (encoder)	a <i>data</i> -side property: whether the query semantically matches a chunk, i.e. whether the answer-bearing content was even available for the memory to encode. Non-obvious because it ignores the model entirely.
<code>first_tok_agree</code>	wrapper vs <code>no_context</code> argmax (first token)	whether the memory changed the base’s answer at all; no change $\Rightarrow$ safe, change $\Rightarrow$ intervention (risky). Borderline-obvious.
<code>ll_argmax_agree_final</code>	layer- ℓ logit-lens argmax = final argmax	decision-crystallisation depth: how early the answer stabilises.

7 Filtering to the non-obvious correlates

The strongest correlates of correctness are unsurprising — the wrapper’s own *confidence* (`top5_mass_w` AUROC 0.653 for `help`; `entropy_0` 0.646 for `noharm`). These are “model-is-sure $\Rightarrow$ model-is-right” and carry little design insight. **Dropping the entire confidence angle** leaves 151 signals; only **11** still clear $|\text{AUROC}_{\text{help}} - 0.5| \geq 0.10$, and they cluster into a single coherent, non-obvious story (Fig. 1):

- **Mid-late-layer wrapper $\leftrightarrow$ base divergence** (`ll_kl_wb_1@25–34`, peak $\ell=30$, $\text{AUROC}_{\text{help}} \approx 0.36$, *inverse*): the more the memory bends the base’s *internal* distribution at the layers where the decision forms, the less likely it helps. Not a confidence signal — a *representational-conflict* signal.
- **Mid-late-layer residual-stream drift** (`drift_1@24–35`, $\text{AUROC}_{\text{help}} \approx 0.40$, *inverse*): the raw magnitude by which the soft prompt perturbs the late hidden state predicts harm — a pure-geometry quantity with no obvious tie to correctness.
- **Layer-localised**: early-layer KL/drift is *not* predictive (mere format perturbation); the signal lives specifically in the decision-forming mid-late band ($\ell \approx 25–34$).

Reading. The design-relevant, non-obvious signal is “*is the memory fighting the base’s own late-layer reasoning?*” (divergence / drift at $\ell \approx 30$), used *inversely*: high conflict $\Rightarrow$ suppress. A naive confidence threshold would miss exactly this, which is what makes it interesting for the gate.

8 The gate target and next steps

Mechanism. Implement the placeholder `gated` combine as an input-conditioned convex mix, optionally *per base layer* ℓ :

$$e'_t = (1 - g_\ell) e_t + g_\ell \tilde{m}_{t,\ell}, \quad g_\ell = \sigma(w_\ell^\top \phi_\ell(\text{signals}) + b_\ell), \quad b_\ell \ll 0, \quad (8)$$

so $g_\ell \approx 0$ at init (wrapper $\approx$ identity, base preserved; cf. LLaMA-Adapter zero-init gating, Flamingo tanh-gating, DISEL input-conditioned LoRA gating). The gate reads the signals the correlation study selects (likely late-layer `conf/ll_conf` for correctness + mid-layer `ll_kl_wb` for compressibility).

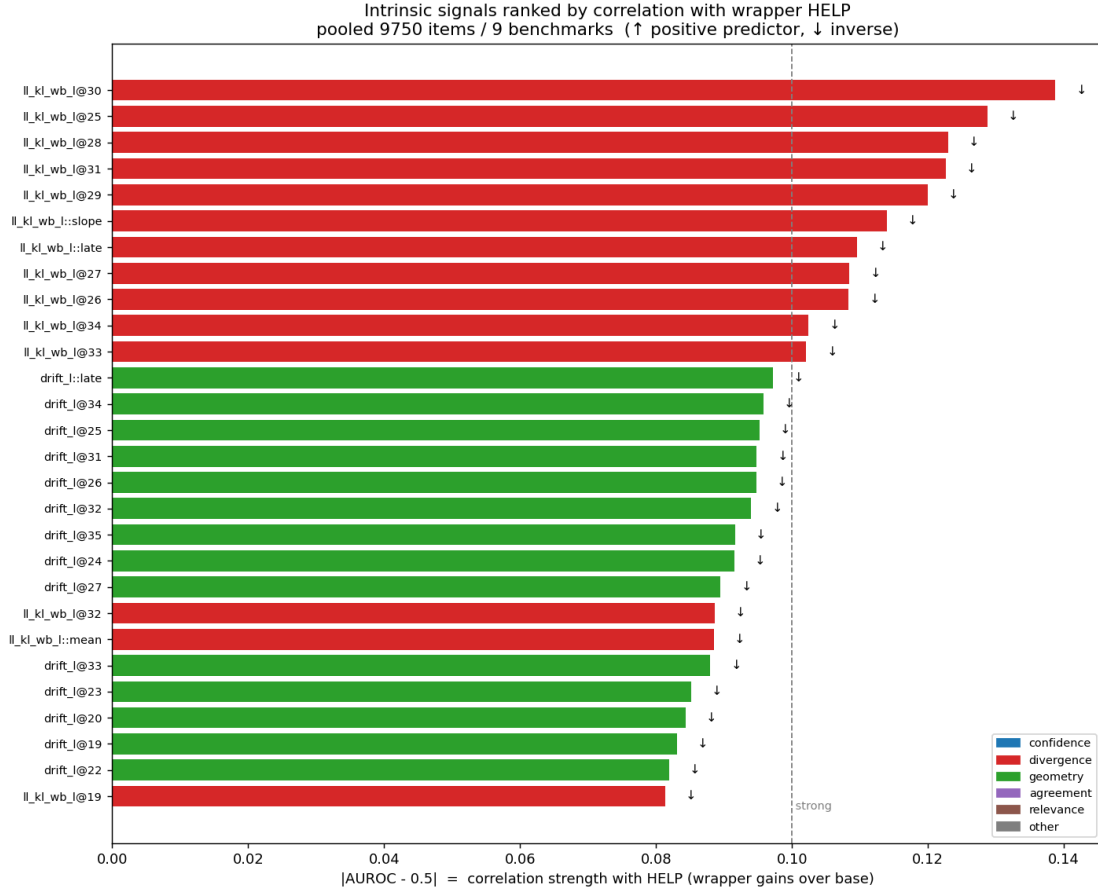


Figure 1: Non-obvious correlates of wrapper `help` (confidence angle removed), pooled 9,750 items over 9 benchmarks, high→low by $|AUROC - 0.5|$ (↓ = inverse predictor). The survivors are mid-late-layer divergence (`ll_kl_wb`) and residual drift (`drift_l`).

Training the gate. On `categorical_niah` + decoy items (irrelevant context $\Rightarrow$ gate should close, target = base `no_context` output) plus a do-no-harm KL regulariser $KL(p_{wrap} || p_{base})$ on held-out OOD-like text. **Eval public-only**; success = Regime-A gain retained *and* `OURS_gated` $\geq$ `no_context` on Regime-B/C.

Pending. (i) Multi-seed bands (9 seeds) to firm up §5. (ii) Multivariate logistic probe for the gate ceiling. (iii) Multi-layer injection (only if the per-layer profile’s payoff over output-only justifies it).

Pointers

- Probe: `mem-embedding/scripts/probe_v2.py` (train+save, native-MC, 37-layer logit-lens/drift).
- Analysis: `scripts/signal_corr.py`; Phase 0 `scripts/signal_study.py`.
- Live ledger: `mem-embedding/summary/matrix.md` (v1 archived in `matrix-archive-2026-06-03.md`).
- Native metrics impl: `llm-infra/src/llm_infra/benchmark_metrics.py`.

- Runs: `runs/pv2-20260604-181915UTC` (seed 42), `runs/pv2-seed*-20260604-*UTC` (multi-seed).